

Program 9

Organize Kubernetes Pods Using Labels and Namespaces

Step 0: Start the minikube

minikube start

```
1rv24mc012_akanksha@akanshasPC:~/Program9$ minikube start
🐳 minikube v1.37.0 on Ubuntu 24.04
🔧 Using the docker driver based on existing profile
👍 Starting "minikube" primary control-plane node in "minikube" cluster
🚚 Pulling base image v0.0.48 ...
🔄 Restarting existing docker container for "minikube" ...
🔧 Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
🔍 Verifying Kubernetes components...
   ■ Using image registry.k8s.io/metrics-server/metrics-server:v0.8.0
   ■ Using image gcr.io/k8s-minikube/storage-provisioner:v5
🌞 Enabled addons: default-storageclass, metrics-server, storage-provisioner
🏠 Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
```

Step 1: Create project folder

mkdir Program-9

cd Program-9

Step 2: Create **namespaces.yaml**

nano namespaces.yaml

Apply it: kubectl apply -f namespaces.yaml

Verify: kubectl get namespaces

```
1rv24mc012_akanksha@akanshasPC:~/Program9$ kubectl apply -f namespaces.yaml
namespace/dev created
namespace/prod created
```

Step 3: Create **pods.yaml**

nano pods.yaml

Apply: `kubectl apply -f pods.yaml`

```
1rv24mc012_akanksha@akanshasPC:~/Program9$ kubectl apply -f pods.yaml
pod/frontend-pod created
pod/backend-pod created
pod/frontend-pod created
pod/backend-pod created
```

Step 4: Verify Pods by Namespace

`kubectl get pods -n dev`

`kubectl get pods -n prod`

```
1rv24mc012_akanksha@akanshasPC:~/Program9$ kubectl get namespaces
NAME                STATUS   AGE
default             Active   23d
dev                 Active   15s
kube-node-lease     Active   23d
kube-public         Active   23d
kube-system         Active   23d
prod                Active   15s
1rv24mc012_akanksha@akanshasPC:~/Program9$ kubectl get pods -n dev
NAME                READY   STATUS              RESTARTS   AGE
backend-pod         0/1     ContainerCreating   0           13s
frontend-pod        0/1     ContainerCreating   0           13s
1rv24mc012_akanksha@akanshasPC:~/Program9$ kubectl get pods -n prod
NAME                READY   STATUS              RESTARTS   AGE
backend-pod         0/1     ContainerCreating   0           22s
frontend-pod        0/1     ContainerCreating   0           22s
```

Step 5: Filter Pods by Label

Frontend pods in **dev**:

`kubectl get pods -n dev -l app=frontend`

Backend pods in **prod**:

`kubectl get pods -n prod -l app=backend`

```
1rv24mc012_akanksha@akanshasPC:~/Program9$ kubectl get pods -n dev -l app=frontend
NAME                READY   STATUS              RESTARTS   AGE
frontend-pod        0/1     ContainerCreating   0           29s
1rv24mc012_akanksha@akanshasPC:~/Program9$ kubectl get pods -n prod -l app=backend
NAME                READY   STATUS              RESTARTS   AGE
backend-pod         0/1     ContainerCreating   0           35s
```

Step 6: Cleanup

```
kubectl delete pods --all -n dev
```

```
kubectl delete pods --all -n prod
```

```
kubectl delete namespace dev
```

```
kubectl delete namespace prod
```